

THE IMPACT OF BRICK AND MORTAR SHOPPING

Advancing our knowledge
of the socioeconomic and
environmental impacts of
shopping centers



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EXECUTIVE SUMMARY

The retail landscape is evolving at a rapid pace, with innovations in retail formats and ways to deliver goods.

Because of these changes, consumers now have more choices to purchase a wide variety of products than ever before. While retailers work hard to deliver convenience and meet customer expectations, shopping behaviors do have socioeconomic and environmental impacts.

Simon undertook this analysis to better understand the economic, social, and environmental impacts of our shopping centers across the company's U.S. portfolio.

The results of this study provides us a baseline from which to measure future progress and help us shape our business strategy towards best serving our retailers and customers.

We are choosing to share this report to advance the conversation—within Simon and at other retailers and the communities in which we operate—about how we can improve our environmental and socioeconomic impact within the ever-evolving retail ecosystem.

The findings in this report highlight the scale of the impact that retail destinations continue to have on local communities:

- The development, construction, and operation of shopping centers **support local and national economies through job creation, small and local business development, and trigger other investments that contribute to the community's local economic development.**
- Shopping malls generate carbon emissions, send waste to landfills, and consume water throughout daily operations, but **present many opportunities to reduce environmental impact.**
- Brick and mortar shopping is not just about consumers purchasing goods and services; **shopping malls also contribute to improvements in quality of life and the communities' wellbeing**, for example our properties work with community members to provide fundraising relating to youth education and development.
- Shopping centers create new destinations for consumers to shop, dine, entertain, and have in-person connections and experiences with friends, family, and community members in the ever-growing age of digitalization.**



INTRODUCTION – REPORT OBJECTIVES AND METHODOLOGY

Why we wrote this report

With a rapid increase in today's omnichannel retail environment, consumers have numerous choices for purchasing products. However, understanding the environmental and socioeconomic impacts of "anything, anywhere, anytime" shopping is not well documented. While retailers work hard to deliver convenience and evolving expectations, shopping behaviors do have environmental and socioeconomic impacts. As the nation's largest owner and operator of shopping malls and a member of communities across the country, it is important for us to understand the impact of our properties.

Simon undertook this analysis to deepen our understanding of the economic, social, and environmental impacts of our shopping centers across our U.S. portfolio. Measuring our full impact will provide valuable insight into our reach, stakeholder engagement, and identify opportunities to enhance our company's corporate responsibility and societal contributions. We sought to answer questions such as:

- How do we positively impact the communities in which we operate?
- How are our properties used, beyond just the purchasing of goods and services, and how do these uses benefit the communities?
- How do we contribute to job creation and economic growth in the communities in which we operate?
- How do our investments in our properties and our supply chain create better economic conditions and stimulate small and local business development?
- How do our environmental initiatives reduce our environmental impact?

In 2016, we published a white paper – *Does shopping behavior impact sustainability?* (the "2016 shopping behavior paper") – where we compared brick-and-mortar mall shopping with online shopping for the same basket of products and identified that online shopping has a 7% larger environmental impact than mall shopping.

With this study, we are expanding upon our 2016 shopping behavior paper's findings on the environmental side by further analyzing the economic and social impacts associated with shopping at a mall. This report provides additional insight to support our business decisions and those of the retail industry more broadly and provides additional content to help shoppers in their shopping decision.

How we conducted the study

We conducted our analysis together with Deloitte Consulting LLP. We developed more than 50 economic and social indicators common for our industry, defined a process for gathering data across our portfolio that would be both high quality and comparable to leading benchmarks, and collected and reviewed data to understand our performance against each indicator. We also compared waste generation and water usage across the entire Simon portfolio with leading industry benchmarks to understand how Simon's performance compares to the U.S. industry as a whole. These indicators and corresponding gathered data will allow us to better understand and measure the socioeconomic and environmental impacts of our properties today and in the long term.

FINDINGS—ENVIRONMENTAL AND SOCIOECONOMIC IMPACTS OF BRICK AND MORTAR SHOPPING

What We Found

What is the true impact of brick and mortar shopping? The findings in this report highlight the scale of the impact that Simon's retail destinations had in the 2017 calendar year on local economies and communities, as well as the national economy.

Key Takeaways

- 5,000 Simon jobs - that's \$372 million in wages.¹
- 311,870 retail jobs across Simon's centers, representing over \$5.9 billion in annual wages.
- \$5 billion in combined property tax payments from Simon and sales tax generated from tenants' sales, delivering significant revenue for state and local governments.
- 100% of Simon shopping malls participate in local community activities and, on average, each of Simon's 203 properties is actively engaged with four community organizations.
- Simon has pledged to donate \$1 million in 2017 and 2018 to Susan G. Komen foundation to support its efforts to end breast cancer.
- \$6,213,025 raised for charity in 2017 from 353 public fundraising events held at Simon properties and attended by 186,740 people.
- Simon's \$5 billion investment in redeveloping its properties over the last five years continues to provide a significant jobs boost to the local economies of the communities it serves.
- More than 30 million visitors enjoyed dedicated play spaces within our shopping malls.
- Our properties send 16% less waste to landfill than the broader industry.
- 100% of our properties have received national 'StormReady' designation by the National Weather Service. StormReady indicates that our properties can act as a safe space during severe weather conditions. Simon is the first REIT to have achieved such status across all its locations.



MEASURING SIMON'S ECONOMIC, SOCIAL, AND ENVIRONMENTAL IMPACT

Determining Our Economic Impacts

The importance of real estate development to the U.S. economy is well established: the \$74.1 billion spent annually on retail construction creates \$101.6 billion in total economic activity for industries and labor up and down the supply chain.² And malls in the U.S. posted their highest ever sales per square foot – \$478 – each year from 2013 to 2015, with adults making an average of 2.1 trips and spending nearly \$162 at malls each month.³

Job Creation

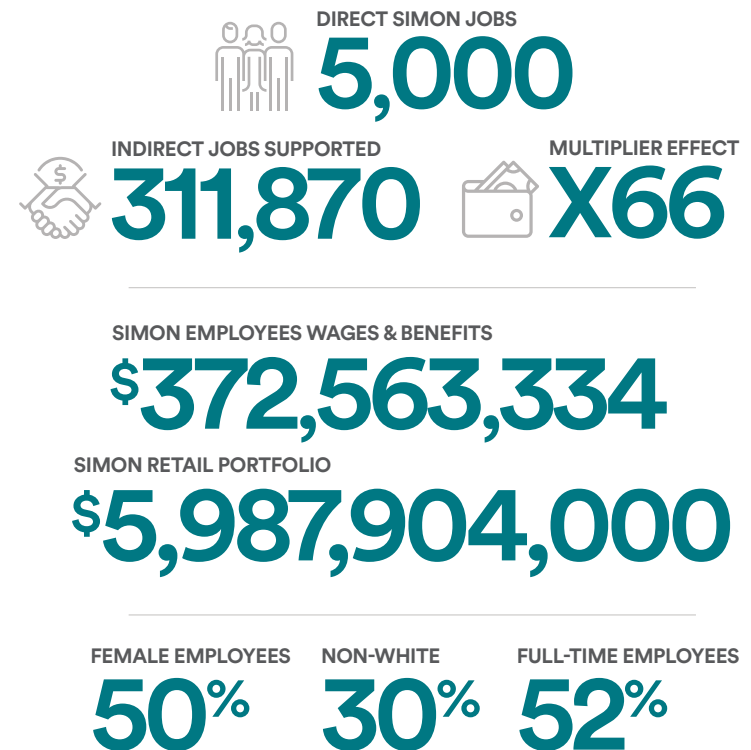
Shopping malls have a multiplier effect on the jobs, wages, taxes, and economic growth sparked in the communities in which they operate. Simon has about 5,000 direct employees (both corporate office and field-based employees). But total tenant or indirect jobs supported by Simon totaled 311,870, meaning that Simon has an employment multiplier effect of more than 66. We support job creation not just for Simon employees, but also for potential tenant employees and suppliers at our properties across the nation. And that does not include jobs at Simon's suppliers such as landscape maintenance workers, janitorial services employees, and the like.

Simon corporate and field direct employees received \$372,563,334 in total wages and benefits. This money is largely injected into the local economies and communities that house our shopping malls through a circular flow of income that provides a multiplier effect on output. The 311,870 tenant jobs supported across Simon's retail portfolio translate to about \$5.9 billion in wages. Wages and benefits can focus on direct purchases (such as product purchases or doctor visits), and be allocated as savings (leading to more business investment in companies) and taxes (which drives investment in municipalities).

Simon supports and encourages diversity across its base of employees and encourages diversity at the uppermost reaches of the company. Approximately 50% of our company is female, and 30% of our employees self-classify as non-white. 52% of our field employees are full-time, a statistic we always seek to strengthen, for we realize the loyalty, employee engagement, and long-term advantages of full-time versus part-time employees.

“Through the development, construction, and ongoing operation of our properties we create jobs, improve infrastructure, and create places where people work, shop, play, and connect. At Simon, we develop and operate first-class properties and positively contribute to both local and national economies, predominantly through job creation, wages and taxes paid, and small and local business development.”

— David Simon, Chairman and CEO Simon



Small and Local Business Development

Many of the tenants that fill Simon properties are large retailers, restaurants, and entertainment groups with names that are familiar to most customers—Macy's, AMC Theatres, California Pizza Kitchen, etc. However, we believe that we also have a unique role in communities and local economies to help boost small and local businesses by assisting them to become our tenants through recruitment and flexible leasing arrangements that allow them to reach new customers and grow their businesses. One such example, currently not available anywhere else, is Simon's new scalable retail platform, first brought to life at our New York-based Roosevelt Field® property and called "The Edit."

The Edit @ Roosevelt Field is a mall-based incubator, where small e-commerce start-ups flourish in a curated space for revolving brands, products, and trends ready to be shopped, shared, and experienced.⁴ While malls have typically been a difficult place for start-ups to gain footholds, The Edit provides a unique, turnkey solution with shorter lease periods, and affords Simon a potentially innovative new option for tenancy and customer attraction.

Our involvement goes well beyond start-ups. We support well-established, local companies and help them expand their businesses by reaching more numerous and more diverse customer bases through our properties. Through innovative leasing terms, we also help these businesses overcome many of the typical barriers to success such as cost- or length-prohibitive leasing terms.

—8,202 local, independent business tenants were part of the Simon portfolio in 2017, providing a unique shopping experience for our customers.⁵

—We also make sure to source materials through local suppliers whenever possible, and spent \$518 million with local suppliers in 2017.⁶

These data points are both notable because dollars spent at locally owned companies tend to create an "economic multiplier" effect whereby such dollars are spread to other goods, services, labor, and charities from local providers—investment in local businesses spurs a recirculation of dollars and reinvestment in the local economy. This means a potentially much larger share of the money spent at a local store will stay in the local economy.

One Salt Lake City-based study of national retail and restaurant chains compared to local retailers and restaurants found that the independent retailers returned a total of 52% of their revenue to the Salt Lake City economy (compared to 14% for the chain retailers), and the local restaurants an average of 79% of their revenue locally (compared to 30% for the chains).⁷



LOCAL, INDEPENDENT TENANTS

8,202

SPENT WITH LOCAL SUPPLIERS

\$518M

CASE STUDY: SUPPORTING PATACHOU RESTAURANT GROUP'S GROWTH

Patachou Restaurant Group—which operates multiple restaurant concepts such as Café Patachou, Petite Chou Bistro and Champagne Bar, and Public Greens Urban Kitchen—is one small local business that Simon has hosted as a tenant and partnered with for growth over the last decade-plus.

In 2006, as we were nearing completion of our new office headquarters in our hometown of Indianapolis, we began searching for a restaurant tenant to serve our employees and others in the downtown area. We were familiar with Patachou restaurants, which at that time were only in neighborhood strip malls, and approached Patachou's owner Martha Hoover to gauge her interest. To Martha, this was a pivotal moment in her company's growth. Up to then, her business had been successful, but once she opened a Café Patachou in our downtown headquarters building, she was able to demonstrate that her company could operate as a regional rather than local player. From her perspective, "While malls are very expensive real estate it is necessary to have mall presence because they are such critical retail locations, especially for customers."

Since the first Café Patachou opened at our headquarters, we have continued to partner with the Patachou Restaurant Group to bring several of their concepts to many of our malls. Martha states, "Now, with the company's presence in Simon's large properties, she has guests who visit the malls from all parts of the country who are now familiar with her restaurants and continue to drive demand for growth and expansion." Simon is proud to have been a part of Patachou Restaurant Group's growth and success as well as that of many other small businesses.



“Simon gave my company more-than-average tenant improvement dollars showing their commitment to getting local businesses at their properties which allowed Patachou to get a real foothold—much more than a small, local landlord could ever do—this kind of investment changes the trajectory of a company.”

— Martha Hoover, CEO, Patachou Restaurant Group



Local Economic Development

Retail is one of the nation's largest non-farm employment sectors and retail development, and ongoing operations are an integral component of any local economy. Studies conducted by the International Council of Shopping Centers (ICSC) have emphasized the importance of shopping centers as a critical source of revenue for local economies, as U.S.-based centers generate nearly \$28 billion in location property taxes each year, and \$167 billion in state and local sales taxes each year.⁸ Property taxes paid by shopping mall owners and sales taxes generated by tenants are significant contributors.

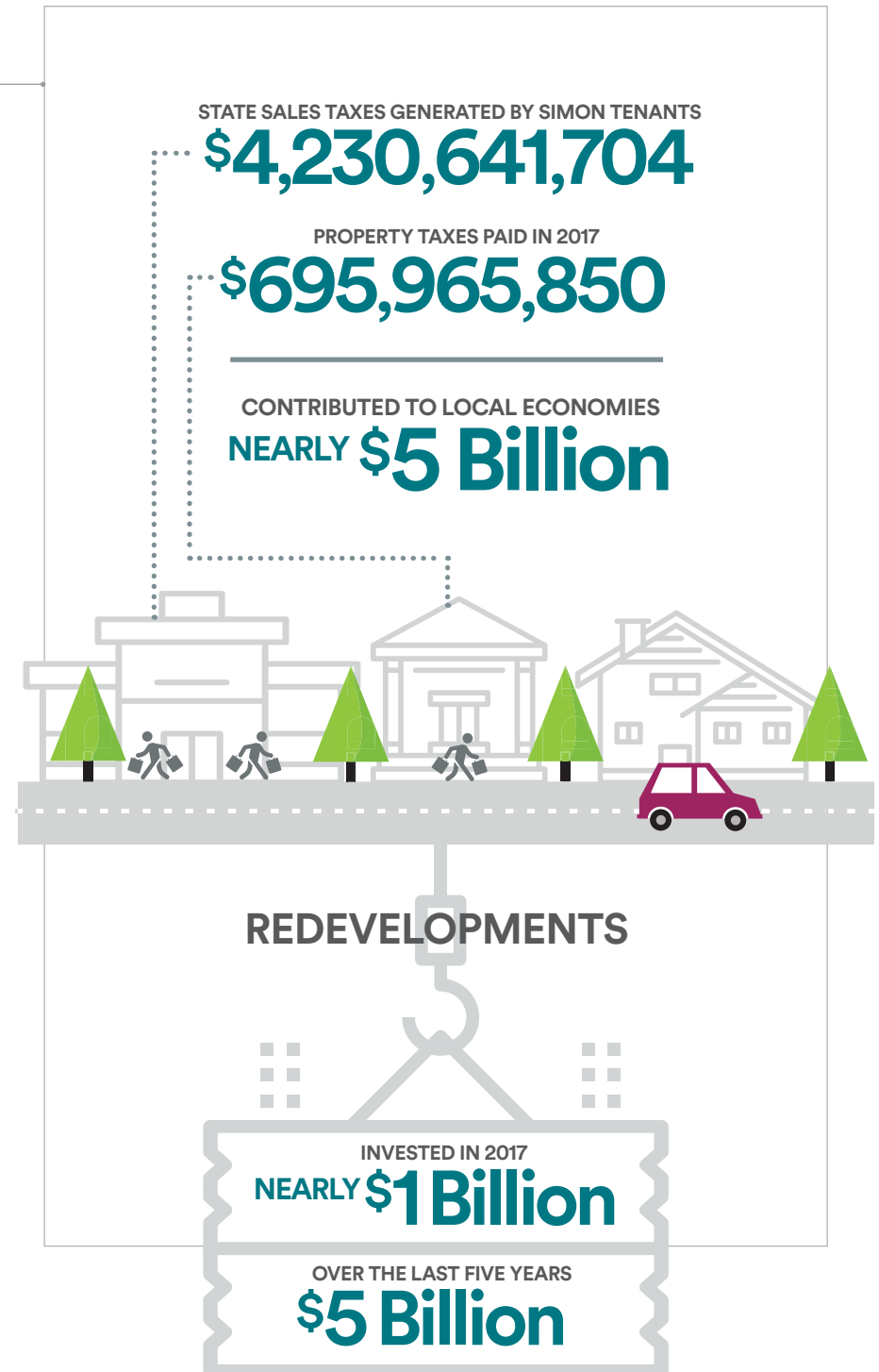
—In 2017, nearly \$5 billion was generated strengthening local economies and communities through the property taxes paid by Simon and sales taxes paid by our tenants.

- \$695,965,850 in property taxes paid by Simon, and
- \$4,230,641,704 in state sales taxes generated by Simon tenants.

Beyond the important role taxes play in the sustainment and growth of local economies, communities also benefit from the revitalization opportunities that shopping malls provide, as malls can drive increased consumer traffic to both local and national stores, restaurants, and other retail establishments, increase the average wage paid in the community, and even enhance the perception and/or beauty of surrounding areas.

—Nearly \$1 billion was invested by Simon in property redevelopment in 2017 to make Simon properties better and provide communities and customers with the best places to shop, dine, connect, and be entertained.

—Over the last five years over \$5 billion was invested by Simon in property redevelopment creating jobs and strengthening our communities.



CASE STUDY: REDEVELOPING SOUTHDALE CENTER IS A WIN-WIN

“Thanks to Simon’s investments, Southdale Center has been a catalyst of \$1 billion of new development in a small suburban community.”

—Scott Neal, Edina City Manager



Southdale Center was the nation’s first indoor regional shopping mall. The center, which opened in 1956, has long been a source of pride for the nearby communities of Edina, Richfield, Bloomington, Eden Prairie, Minneapolis, and St. Louis Park. However, over its lifespan, the mall has hit tough skids which has been difficult for the community due to its pride in the property.

When Scott Neal, the current City Manager, arrived in Edina in 2010, he witnessed concern not just about the shopping center but also about what its decline would mean for the community. As Neal states, “The health of the shopping center has direct impact on the economic impact of properties around it – if we could improve the health of Southdale Center, we knew it would have a significant economic impact on not just the mall itself but all surrounding properties.” Shortly after that, Simon Property Group acquired Southdale Center and began extensive renovations.

Since the acquisition, Simon has invested heavily in renovating the property and bringing in new tenants. More remarkably though, the rejuvenation of the mall has spurred extensive growth in the surrounding community. The growth of Southdale Center and the surrounding properties has been extensive:

- New Development: Over 1,700 housing units built or under construction, and an additional 550 units approved and waiting for construction.
- Community Investment: Over \$900 million in new expected market value (EMV) from new construction, and an additional \$250 million approved in new EMV awaiting construction with \$400 million in new EMV under consideration.

As an example, Simon has partnered with the city to redevelop the vast, underutilized parking area—the remnant of the original 1950’s design when customers relied more on cars rather than public transportation to access the mall—into a 232-unit apartment complex called One Southdale Place. Not only has the estimated market value of this land increased by more than tenfold from \$4.1 million to \$53.4 million since it was redeveloped, but it is also generating six times more local taxes (from \$148,684 in 2013 to \$886,171 in 2017). This kind of growth all around Southdale Center benefits the community in countless ways, much beyond the services provided just by Simon’s Southdale Center.

Measuring Our Social Impacts

The influence of shopping malls goes beyond the environment. Shopping centers are an integral part of the economic and social fabric of communities.⁹ This section highlights some of the socioeconomic effects of Simon's shopping malls, including the benefits they offer in the form of accessible spaces, fundraising, and youth development opportunities.

Simon Shopping Centers Support Our Communities

Shopping malls provide spaces for socialization, play, exercise, education, and even gathering locations during community disasters. Other activities our shopping centers afford today's time-crunched consumer include religious centers, postal branches, libraries, museums, and offices. Shopping centers create new destinations for consumers to shop, dine, entertain, and have in-person connections and experiences with friends, family, and community members in the ever-growing age of digitalization.

In 2017, 2,827,295 square feet of Simon-finished common area property was utilized for community purposes (e.g., areas for fundraising activities, etc.).

—Contained within these community-specific spaces are **128,000 square feet of dedicated play spaces that hosted over 30 million visitors** throughout the course of the year. The importance of play for a child's healthy development is well understood and accepted, but public areas for recreation are not always available or accessible. Our shopping malls provide free and accessible environments for play both outdoors and indoors.



DEDICATED TO PLAY SPACES

128,000 SF

VISITORS HOSTED

30 Million+

MEASURING SIMON'S ECONOMIC, SOCIAL, AND ENVIRONMENTAL IMPACT

—1,849,862 total mall walkers across all U.S.-based Simon properties in 2017—
shopping and exercise, together!

Walking is a generally safe and comfortable form of exercise. However, walking outside is not always possible. The Office of Disease Prevention and Health Promotion has highlighted that mall walking “programs can provide safe, convenient, and comfortable places for residents to be physically active and make social connections. Using existing malls to provide spaces, where people can walk regularly, has the potential to address barriers some people face in getting regular physical activity.”¹⁰

Our malls provide free, accessible, pedestrian-friendly, and climate-controlled environments. The Westchester mall in New York has even hosted mall walking every Tuesday and Friday for more than a decade!

—100% of Simon properties have received “StormReady” designation, providing safe spaces and shelters during severe weather.

StormReady is a national, voluntary community preparedness program that supports commercial gathering sites in preparing for severe storms. With guidance provided from the local National Weather Service Office, state and local emergency managers, and the media, the program provides communities with the communication and safety skills needed to save lives and property. Designation means the property has demonstrated readiness for all types of severe weather. We are proud that all of our properties are StormReady, and that we are the first REIT to have achieved such status.

—63 disaster relief events, such as food drives, were held at Simon properties in 2017. Beyond offering shopping, entertainment, and dining options, our properties have uniquely served as a gathering place and collection point.

In 2017 during Hurricane Irma, some of Simon properties in Miami, including Dadeland Mall, The Falls, Miami International Mall, and Coral Square served as drop-off locations for the Red Cross to assist those affected by the fifth-most damaging hurricane on record. And during the 2017 Northern California wildfires, our Petaluma Village Premium Outlets® hosted a goods collection drive through which all proceeds went directly to victims and fire relief efforts, and the property’s parking lot was used as a staging area for approximately 2,000 evacuees while they waited for shelters to open in the surrounding area.



TOTAL MALL WALKERS AT SIMON PROPERTIES

1,849,862



STORMREADY DESIGNATION

100%



CASE STUDY: PARTNERING WITH PETA FOR HUMAN AND ANIMAL WELFARE EDUCATION

Simon has engaged in a unique partnership with People for the Ethical Treatment of Animals (PETA) to raise awareness about the danger of leaving small children and animals in cars. For the last two years, Simon has posted prominent signage in parking lots and at property entrances to remind customers to check their cars and ensure that they haven't forgotten any passengers.

Getting the word out: While difficult to quantify the impact of these efforts, PETA's Manager for Corporate Affairs and Senior Corporate Liaison, Stephanie Shaw has stated that "Signage in malls is one of the most effective methods of getting the word out, to help remind people never to leave *anyone* in their cars, and Simon certainly raised public awareness while opening the floodgates of getting other property owners to agree to post similar messages."

Continuing partnership: "This work is the continuation of a longstanding partnership that has led to significant advancements in animal welfare," said Stephanie Shaw. For example, in 2011, Simon created a new policy based on its work with PETA not to allow circuses that use wild animals on its properties due to the inherent animal welfare and human safety concerns, and now more than 650 malls owned by multiple companies nationwide have followed suit. And in 2013, Simon again took a leadership stance on animal welfare when it banned the sale of sugar gliders, which are small, tree-dwelling marsupials, on its properties, leading to there now being 1,300 malls in the U.S. where 'pocket pets' are prohibited.



CIRCLE OF
RESPECT™

SHOP SAFE

**Never leave children or pets
unattended in vehicles.**

Lock your car.

Hide your valuables.

Take your keys.

Simon is a proud sponsor of the National Crime Prevention Council (NCPC). The Circle of Respect is a registered trademark of NCPC (www.ncpc.org).



Fundraising Support

Simon shopping malls fulfill a community need by providing the space and support required for charitable fundraising. We meet those needs through a variety of ways, but most notably through national and local community partnerships and fundraising events held at our properties. Examples are listed below.

—The national community partnership between Simon and Susan G. Komen. Simon has long supported the Susan G. Komen foundation and its efforts to reduce incidences of breast cancer deaths. We have pledged to donate \$1 million in both 2017 and 2018, and more than 180 Simon Malls,® The Mills,® and Simon Premium Outlets® nationwide participated in a range of activities in support of the foundation during 2017. In partnership with renowned yoga brand Manduka during Mother’s Day weekend of 2017, we participated in project OM—a one-million-person yoga event that united individuals, families, and the collective yoga community to raise funds to support Susan G. Komen and its Bold Goal of reducing breast cancer deaths by 50% in the next decade.

- 4 local community organization partnerships held on average by each of our properties.
From national organizations such as the Salvation Army, Red Cross, Girl Scouts and Autism Speaks to smaller, regional-based organizations, each of our more than 200 mall properties maintains an average of four partnerships with organizations in their local communities in any given year.
- 353 fundraising events (such as charity runs/walks and holiday-focused fundraisers) and other non-fundraising events (such as blood drives and diaper banks) held at Simon properties in 2017.
186,740 people attended these 353 events throughout the year, and this amounts to an average of nearly 530 attendees per event! But this only tells half the story, as our staff also contributed 3,811 hours, or nearly 11 hours per staff member, to each event.
- \$6,213,025 was raised by community members across these various events and collected at Simon properties in 2017.



CASE STUDY: CARING BUNNY AND CARING SANTA PROGRAM WITH AUTISM SPEAKS

For many of us, our holiday memories include trips to visit Santa or the Easter Bunny, with annual photos that serve as reminders of childhood holiday traditions. As community gathering spaces, malls have hosted these holiday photo events for generations. For some of our guests, though, this tradition feels far out of reach—crowded malls, long lines, and proximity to strangers can be too overwhelming for children with special needs.

At Simon, we have partnered with Autism Speaks to host our Caring Bunny and Caring Santa program. Caring Bunny and Caring Santa events allow children with special needs and their families the opportunity to visit with the Easter Bunny and Santa in a subdued, quiet environment. These families want to enjoy the same traditional experiences that most other children experience. Because of that, we take special care to create an appropriate and fun environment for these kids. Typically, more than 100 of our malls participate in the program each holiday season.

Most Caring Bunny and Caring Santa events are held on Sunday mornings during the holiday seasons for a few hours before the mall opening. During these events, our mall teams make specific adjustments to the environment of the mall such as turning off music, fountains, etc. to support the sensory, physical, and other developmental needs of children of all abilities. Our guests regularly tell us how special it is for their families to be able to participate in these longstanding holiday traditions—for Simon, it's just one small way that we can support inclusivity in our malls and communities.¹¹



“This was our only way. We wouldn’t get traditional Santa pictures otherwise. For years we didn’t because it was too much. So to be able to do something that’s ‘normal,’ like have a traditional photo and a traditional childhood memory, is great,” said the mother of one of our guests. “Now we can have an actual Christmas picture, and our son feels safe.”

Youth Development

Spending time at the mall is part of any American's childhood years. And this is not just because of the social aspects these properties provide. Malls also provide educational opportunities.

We support Simon Youth Foundation (SYF), which was established in 1998 by a group of Simon employees in Indianapolis. Seeing far too many kids in malls instead of classrooms during school hours, Simon employees discussed bringing the classrooms to the mall, which formed the basis of the Foundation.

SYF's mission is to help public school systems reach at-risk kids who have dropped out or are on the verge of dropping out of their traditional high school by establishing Simon Youth Academies in Simon Malls that offer flexible schedules and small class sizes making learning accessible to students who struggle to connect with the material in a traditional classroom because of their circumstances.

- 2,735 students are enrolled in Simon Youth Academies.
- 43,714 square feet of space dedicated to 11 SYF Academies in 2017 (programs typically occur within Simon properties).
- 15,660 students graduated from the SYF program since 1998, with a graduation rate of 90%.
- 5,000 students supported with over \$17 million in scholarships since SYF's inception in 1998.



“ [Judson Learning Academy at Rolling Oaks Mall is] a normal high school thinking outside the box on how to educate different people. The regular classroom just wasn't for me...if it wasn't for the Judson Learning Academy, I probably would have gave up.”

— Timothy Wilson, former Mayor of Kirby, Texas



OUR ENVIRONMENTAL IMPACT

GHG Emissions

Our 2016 shopping behavior white paper (*figure 1*) identified the greenhouse gas (GHG) emissions associated with mall and online shopping and established that the choices customers make regarding how they buy products and how they utilize product return options have clear impacts on the environment. A representative shopping basket of four products was considered for both mall and online scenarios.¹²

The analysis examined five stages across the product life cycle: Delivery and Logistics; Shopping; Packaging; Customer Travel or Last Mile Delivery; and Returns. The research focused on the life cycle phases that have differences between mall and online shopping, and considered how customers utilize each of the services within each life cycle phase. We found that the main contributors that affect the level of GHG emissions in either shopping experience include transportation fuels, building energy usage, and packaging differences.

Results highlighted that online shopping has a 7% larger environmental impact than mall shopping if shoppers purchased the same number of products via a brick-and-mortar mall as online.¹³

In an age when consumers are increasingly demanding same-day or fast delivery, which requires more resources such as fuel to fulfill, the negative impact of online shopping is likely to worsen even more. These last mile deliveries are causing traffic congestion, noise pollution and stressing our

infrastructure. Put simply, the choices customers make regarding how they buy products and how they utilize product return options have clear impacts on the environmental footprint.

And did you know that on average 33% of online purchases are returned versus 7% of brick-and-mortar purchases?¹⁴ When you take into account this frequency, plus the transportation and use of any electronics needed to support the return, for a similar basket of goods there is an additional footprint of 157 kg GHG emissions for online returns as compared to mall returns.

With this current white paper, we expanded the Life Cycle Assessment (LCA) to include waste and water, so that we may consider the use and impacts of additional resources beyond just the energy and emissions detailed in the 2016 paper. Against a similar basket of goods, we found that the majority of retail goods' waste and water footprints is in the raw material and manufacturing stages, not in the logistics and distribution, customer shopping/interface, and product delivery phases—which were the focus of the previous 2016 paper's LCA. Therefore, we concluded that there is no discerning difference or significant factors are impacting the waste or water impacts of mall and online shopping. So, we decided to pursue a portfolio approach to compare Simon's total water use and waste across our property portfolio to industry benchmarks to obtain a better understanding of our performance relative to peers.

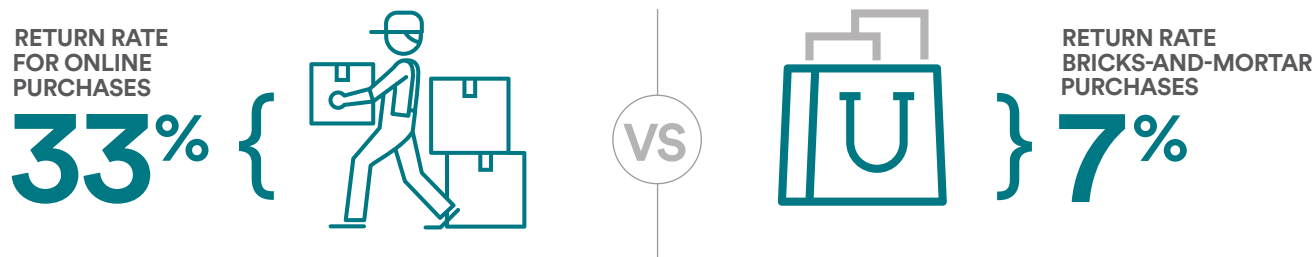


Figure 1

PHASE	FUEL EMISSIONS	ENERGY EMISSIONS	PACKAGING EMISSIONS	TOTAL EMISSION	% OF RESPECTIVE TOTALS
LOGISTICS & DISTRIBUTION					
MALL	6,197	1,616		7,814	21%
ONLINE	10,951			10,951	27%
CUSTOMER SHOPPING					
MALL		10,264		10,264	27%
ONLINE		1,976		1,976	5%
PRODUCT DELIVERY					
MALL	19,325		308	19,633	52%
ONLINE	24,847	163	2,359	27,369	68%
TOTAL FOR MALL	25,523	11,880	308	37,710	100%
TOTAL FOR ONLINE	35,798	2,139	2,359	40,295	100%

This shows emissions from malls 7% lower than online 2,585 <=Difference



6.2 million miles driven by an average US passenger car

THE IMPACT DIFFERENCE IS THE SAME AS:



68,000 incandescent bulbs replaced with CFLs



[Click here to view full report](#)

Waste

The average American generates 4.48 pounds of trash per person per day.¹⁵ The waste consists of various items thrown away from residences, institutional facilities, and commercial properties, of which shopping malls are a component. Waste generated at a shopping mall is pervasive: From packaging- and operational-related waste to food waste and construction-related waste, trash is generated through many avenues from both retailers and consumers. As part of this analysis, we looked at our waste generated on a per-square-foot basis, which enabled comparison to industry benchmarks and provided a better understanding of our performance relative to peers.

We compared our 2016 waste data (comprised of waste generated from our retail tenants and consumers) to national benchmarks for the shopping center industry from the Urban Land Institute.^{16,17} We found that our waste-to-landfill intensity (or the pounds of waste we sent to landfill per square foot of property) is 16% less than our peers and that our diverted waste (or the waste sent for recycling or waste-to-energy rather than landfill) is 102% that of the industry. In other words, Simon is sending less waste to landfill and recycling more waste than the industry.

Simon is continuously looking for opportunities to increase recycling and minimize waste sent to landfills. Our ability to offer recycling options depends on various factors, such as the local recycling infrastructure, the availability of recyclable materials collection services, and tenant participation in recycling programs, because our tenants generate the vast majority of the waste produced at our properties.

We maintain cardboard recycling at all properties—and recycled the equivalent of 340 Olympic-sized swimming pools, or more than 125,000 tons, of cardboard in 2016—and have single-stream recycling at more than 70 of our centers. We have already completed our goals of reusing or recycling 70% of construction waste generated during major renovations. And we have introduced plastic thin film recycling at ten centers and food waste recycling programs at nine centers.

Water

Given that the EPA's Water Sense program has found that 40 out of 50 U.S. states' water managers expect water shortages under average conditions in some portion of their states over the next decade, responsible water management is crucial for Simon's ongoing business operations. Water is used in shopping malls in restrooms, restaurants, cooling systems, and irrigation. Typically, this water is then discharged to the local sewer system.

Simon's water performance was compared to national benchmarks for the shopping center industry from the Urban Land Institute's *Greenprint Performance Report*.^{18,19} Simon's total water intensity of 17.91 gallons per square foot is about 1.2% higher than industry intensity, showing our water usage is on par with the broader industry.

We are working towards responsible water management and reducing water consumption across our portfolio. We employ leading practices, such as smart irrigation technologies (including local drip-irrigation systems and high-efficiency sprinkler systems), smart restroom technologies (including low-flow fixtures and toilets, waterless urinals, and faucet aerators), improving efficiencies in building cooling towers, and partnering with our tenants where possible to manage our water footprint.

We regularly conduct water risk assessments (e.g., leveraging World Resources Institute's Aqueduct mapping tool) to better understand our water risk exposure and make appropriate management and investment decisions. Finally, we continue to work towards our broader goal of improving the water efficiency of our portfolio 20% by 2025.

Shopping malls use energy and water and produce GHG emissions, waste, and wastewater from daily operations. But so does the whole retail industry, and there are success stories evident every day around reducing energy usage and GHG emissions, waste generation, and water consumption.

**At Simon, we constantly work to reduce our environmental impact—
read more in our sustainability report at simon.com/sustainability.**

CONCLUSION AND NEXT STEPS

What's Next?

By analyzing the broader, socioeconomic perspective, Simon has demonstrated that shopping centers have many positive impacts in their communities. Moreover, this report highlights how our shopping centers provide benefits from an economic, social, and environmental standpoint.

Malls provide convenience not only through myriad retail and dining options for consumers, but also acting as “one stop” locations that provide additional destinations and services that help people save time, reduce their commutes, and reduce GHG emissions from transportation. And they offer a benefit, enabling socialization, play, exercise, fundraising, and even education through the diverse, community-oriented spaces they provide. Through supporting job creation, reinvesting in communities through circular flows of income to local purchases, and engaging and partnering with small and local businesses and suppliers, shopping centers also have a notable impact on both local and national communities and economies. Although shopping centers do use energy and water, and generate GHG emissions and waste, our properties also present an opportunity to reduce this impact—whether through innovative energy efficiency practices, waste reduction campaigns, use of water conserving technologies, or similar—while simultaneously highlighting opportunities for our retail partners and consumers to reduce their own environmental impact, as well.

In summary, Simon continues to strategically invest in and improve its portfolio of properties, and these actions will have positive impacts on the communities in which we operate.



APPENDIX

Detailed Methodology

We have developed a data-driven methodology to support this study using Simon's proprietary data to quantify how the development, construction, operation, and management of our properties contribute to their environmental footprints, create jobs, trigger economic development and community development, and establish new destinations for communities in which we operate. We developed economic, social, and environmental indicators common for our industry, and gathered data to understand our performance against each indicator.

Socioeconomic Analysis

To conduct the socioeconomic analysis, a "bottom up" approach was applied whereby a comprehensive set of socioeconomic factors was considered and analyzed through a number of steps as described below.

1. Build a comprehensive set of potential socioeconomic indicators by benchmarking peer companies, sustainability leaders from other industries, and globally accepted reporting guidelines (Global Reporting Initiative (GRI), Global Real Estate Sustainability Benchmark (GRESB), and United Nations Global Compact (UNGC)) to understand the landscape of socioeconomic indicators.
2. Prioritize indicators to track and report on by examining them to understand each indicator's feasibility and utility.
 - a. The following principles were established to determine whether to include an indicator in the study, and a list of 50 indicators was subsequently developed.
 - i. Is it a metric that peer companies report on?
 - ii. Is it a metric that external stakeholders (investors, NGOs, etc.) do expect or might expect Simon to report on?
 - iii. Does the metric help Simon understand its impact?

3. Define process for collecting (e.g., through conversations, emails, surveys, etc.) and reviewing (i.e., assuring high quality) data.
 - a. Only 2017 data was collected for the socioeconomic indicators to ensure common parameters and comparability of data sets.
4. Gather and analyze data to test hypotheses and provide insights regarding Simon's performance against indicators.
 - a. All data was collected and quality assurance was performed before aggregating data into the appropriate metrics.

Environmental Analysis

To fully understand the environmental impact of mall shopping, this study was designed to look beyond energy use and greenhouse gas emissions to also identify the impact of shopping on waste and water. An initial analysis found that a comparison of mall and online shopping waste and water footprints would not yield meaningful results, that there were no significant factors impacting waste or water impacts that differed between online and mall shopping. Given this lack of insight from a LCA approach, 2016 data submitted as part of Simon's 2017 GRESB response was leveraged in order to help us better understand our impacts on waste and water.

To further understand this data and how Simon's waste and water intensities compared to the rest of the REIT industry, we researched industry organizations and environmental groups to identify credible industry benchmark data. We focused on data from the 2017 Urban Land Institute Greenprint Performance Report ("ULI" – Volume 8). This report summarizes collected 2016 performance data on 8,684 properties across 1.9 billion square feet of building area in 28 countries. For the purposes of benchmarking Simon's performance, we then identified an abbreviated but comparative data set from this report consisting of shopping centers only: 32 centers that responded with waste data, and 122 centers that responded with water data.

Footnotes

- (1) \$12/h wage at 40h weeks and 10-month full employment annually.
- (2) ICSC, “Consumers See a Bright Future in Malls,” March 2, 2018. Accessible at https://www.icsc.org/uploads/t07-subpage/Consumers_See_a_Bright_Future_in_Malls.pdf.
- (3) Simon analysis.
- (4) For more information, refer to <https://www.simon.com/the-edit>.
- (5) Simon’s definition of a local, independent business is a business for which the location under question is the first location for that business in the market or is a start-up, and the business is locally owned and operated (not operated or funded by a large national brand or chain).
- (6) Suppliers in categories including utilities; security; janitorial, food court, and landscaping; administration; and repairs and maintenance.
- (7) Civic Economics, “Indie Impact Study Series: Salt Lake City, Utah” August 2012. Accessible at <http://www.localfirst.org/images/stories/SLC-Final-Impact-Study-Series.pdf>.
- (8) ICSC, “The Impact of Shopping Centers on the U.S. Economy.” Accessible at <https://www.icsc.org/uploads/t07-subpage/US-Economic-Impact-2018.pdf>.
- (9) ICSC, “2014 Economic Impact of Shopping Centers.” Accessible at <https://www.icsc.org/uploads/default/2014-Economic-Impact-Kit.pdf>.
- (10) David Brown, Office of Disease Prevention and Health Promotion, “Mall Walking Programs Can Help Promote Physical Activity and Health,” March 9, 2016, accessible at <https://health.gov/news/blog-bayw/2016/03/mall-walking-programs-can-help-promote-physical/>.
- (11) Quote provided by The Today Show. Accessible at <https://www.today.com/parents/mall-santa-goes-extra-mile-boy-autism-they-just-bonded-t59651>.
- (12) Four products included a women’s top, women’s shoes, coffee maker, and set of six wine glasses.
- (13) Simon, “Does shopping behavior impact sustainability?” March 2016.
- (14) Banjo, Shelly. “Rampant Returns Plague E-Retailers.” WSJ. *Wall Street Journal*, 22 Dec. 2013. Web. 03 Nov. 2015
- (15) U.S. EPA, “Advancing Sustainable Materials Management: 2015 Fact Sheet,” released July 2018. Accessible at https://www.epa.gov/sites/production/files/2018-07/documents/2015_smm_msw_factsheet_07242018_fnl_508_002.pdf.
- (16) Data is for 2016, as this was the most recent full year of data available, for both Simon and industry.
- (17) Urban Land Institute (ULI) *Greenprint Performance Report: Volume 8* (2017). Includes 2016 data on 8,684 properties across 1.9 billion square feet of building area in 28 countries. The data set included 32 shopping centers that provided waste performance data which was used for the benchmarking conducted in this report.
- (18) Data is for 2016, as this was the most recent full year of data available, for both Simon and industry.
- (19) Urban Land Institute (ULI) *Greenprint Performance Report: Volume 8* (2017). Full report includes 2016 performance data on 8,684 properties across 1.9 billion square feet of building area in 28 countries. A comparative data of 122 responding shopping centers (with water data) was identified for purposes of benchmarking to Simon performance.